

Course Path

STAT 109 — Summer Session 1, 2026 (online, async)

This page shows the recommended path through the course during the five-week summer session from June 1 to July 2, 2026. You may work at your own pace within each module window, but the modules should be done in the order shown. Check [Canvas](#) to turn in work, take quizzes, see grades, and join discussions.

Reference sheets ([methods map](#), [R reference](#)) summarize all course content. All readings also live on the [lectures](#) index if you want the full list.

The Supplemental (optional) column links extra textbook chapters (Vu & Harrington, free online) and videos/tutorials that match our Colab + `{tidyverse}` workflow — see [Resources](#). None of these are required for the module project unless a row says otherwise.

How the course is graded

Component	Weight	What counts
Class participation	5%	Class Chatter (one per module, 0–5) + Q+A Discussion on Canvas
Quizzes	25%	All Canvas quizzes (Quiz 0/0B, Quiz A, Quiz B, synthesis per module) · best of 4 attempts per quiz · open notes
R coding exercise labs	20%	Submit exercise lab each module · graded on completion
Projects 1–5	50%	One-page summary + runnable R notebook each module
Total	100%	

At a glance

Module	Dates	Project due (recommended)
0 — Welcome	Jun 1–2	—
1 — Testing a Claim	Jun 2–8	Mon Jun 8
2 — Estimating the Truth	Jun 9–15	Mon Jun 15
3 — Discovering Patterns	Jun 16–22	Mon Jun 22
4 — Designing Fair Comparisons	Jun 23–29	Mon Jun 29
5 — Looking Ahead	Jun 30–Jul 2	Thu Jul 2

Canvas due dates are recommendations for pacing; all work to be graded must be submitted by 11:59 pm PST July 2, 2026.

Module 0 — Start here

Step	Materials	Supplemental (optional)
Read 0	Module 0 Read 0	—
Read	Syllabus · Course path · Resources	—
Quiz	Quiz 0 (Start Here) · Quiz 0B (Question Ideas)	—

Step	Materials	Supplemental (optional)
Lab	Demo lab · Exercise lab	—
Canvas	Module 0 discussion (introduce yourself)	—
Supplemental pack	—	Colab tutorial · Posit: Getting Started with R · Textbook (free PDF) · Posit cheatsheets

Module 1 — Testing a Claim

Is this observation surprising?

Step	Materials	Supplemental (optional)
Read 0	Module 1 Read 0	—
Read	Module 1 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 1 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 1 demo · Conditional demo	—
Exercise lab	Module 1 exercise lab	—
R reference	R quick reference	—
Milestone	Project 1 milestone · notebook	—
Synthesis	Module 1 synthesis	—
Project	Project 1	—
Supplemental pack	—	Textbook Ch 2 — Probability · Ch 3.2 — Binomial · Khan: binomial distribution · Khan: visualizing binomial · Generative AI (syllabus policy)

Module 2 — Estimating the Truth

Building high-quality estimates from samples.

Step	Materials	Supplemental (optional)
Read 0	Module 2 Read 0	—
Read	Module 2 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 2 Read 2	Textbook Ch 4.2 — Confidence intervals
Quiz B	Concept Check	—
Demo lab	Module 2 demo	—
Exercise lab	Module 2 exercise lab	—
Milestone	Project 2 milestone	—
Synthesis	Module 2 synthesis	—
Project	Project 2	—

Step	Materials	Supplemental (optional)
Supplemental pack	—	Textbook Ch 3.3 — Normal · Ch 4.1–4.2 — Variability & CIs · Khan: confidence intervals · Khan: CI simulation

Module 3 — Discovering Patterns

Describing a dataset with graphs and summaries.

Step	Materials	Supplemental (optional)
Read 0	Module 3 Read 0	—
Read	Module 3 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 3 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 3 demo	—
Exercise lab	Module 3 exercise lab	—
Milestone	Canvas Module 3	—
Synthesis	Module 3 synthesis	—
Project	Project 3	—
Supplemental pack	—	Textbook Ch 1 — Intro to data (§1.4–1.7) · R4DS: Data visualization · Posit: ggplot2 video

Module 4 — Designing Fair Comparisons

Comparing two groups responsibly.

Step	Materials	Supplemental (optional)
Read 0	Module 4 Read 0	—
Read	Module 4 Read 1	—
Quiz A	Def & Notation	—
Read more	Module 4 Read 2	—
Quiz B	Concept Check	—
Demo lab	Module 4 demo	—
Exercise lab	Module 4 exercise lab	—
Milestone	Project 4 milestone	—
Synthesis	Module 4 synthesis	—
Project	Project 4	—
Supplemental pack	—	Textbook Ch 5.3 — Two-sample tests · Ch 8.1–8.2 — Proportions · R4DS: Data transformation

Module 5 — Looking Ahead

Using data to make predictions.

Step	Materials	Supplemental (optional)
Read 0	Module 5 Read 0	—
Read	Module 5 Read 1	Regression line applet
Quiz A	Def & Notation	—
Read more	Module 5 Read 2	Penn State: prediction intervals
Quiz B	Concept Check	—
Demo lab	Module 5 demo	—
Exercise lab	Module 5 exercise lab	—
Synthesis	Module 5 synthesis	—
Milestone	Canvas Module 5	—
Project	Project 5 · Portfolio package due Thu Jul 2	—
Supplemental pack	—	Textbook Ch 6 — Simple linear regression · Khan: regression line example · Penn State: prediction intervals